

Question:

The \$y\$-intercepts, \$P\$ and \$Q\$, of two perpendicular lines intersecting at the point \$A(6,8)\$ have a sum of zero. What is the area of \$\triangle APQ\$?

\$(A)\$ 45 \$\quad\$ \$(B)\$ 48 \$\quad\$ \$(C)\$ 54 \$\quad\$ \$(D)\$ 60 \$\quad\$ \$(E)\$ 72

Reasoning + Answer:

<think>

1. Given that the lines are perpendicular and intersect at point $A(6, 8)$, let's denote the slope of one line by m_1 and the slope of the other line by m_2 . Since the lines are perpendicular, the product of their slopes is -1 , i.e., $m_1 \cdot m_2 = -1$.

2. The equation of a line with slope m passing through point (x_1, y_1) is given by $y - y_1 = m(x - x_1)$. For line 1 with slope m_1 passing through $A(6, 8)$, the equation becomes:

$$y - 8 = m_1(x - 6)$$

Simplifying, we get:

$$y = m_1(x - 6) + 8 = m_1x - 6m_1 + 8$$

3. For line 2 with slope m_2 passing through $A(6, 8)$, the equation becomes:

$$y - 8 = m_2(x - 6)$$

Simplifying, we get:

$$y = m_2(x - 6) + 8 = m_2x - 6m_2 + 8$$

4. Since the y -intercepts P and Q of the lines intersecting at point A have a sum of zero, let $P = (0, p)$ and $Q = (0, -p)$. This implies that the lines pass through $(0, p)$ and $(0, -p)$, respectively.

5. The coordinates of P and Q can be expressed as:

$$P = (0, p) \implies y = p$$
$$Q = (0, -p) \implies y = -p$$

6. Given that the lines intersect at $A(6, 8)$, the coordinates of A must satisfy both lines' equations:

$$8 = m_1(6) - 6m_1 + 8 \quad \text{(equation for line 1)}$$
$$8 = m_2(6) - 6m_2 + 8 \quad \text{(equation for line 2)}$$

7. Simplifying these, we get:

$$8 = 6m_1 \quad \text{(from line 1)}$$
$$8 = 6m_2 \quad \text{(from line 2)}$$

8. Solving for m_1 and m_2 :

$$m_1 = \frac{8}{6} = \frac{4}{3}, \quad m_2 = \frac{8}{6} = \frac{4}{3}$$

9. Therefore, the slopes of the lines are $m_1 = \frac{4}{3}$ and $m_2 = -\frac{4}{3}$.

10. Now, we need to find the area of triangle APQ . The base PQ is along the y -axis and has length $2p$. The height is the distance from A to the y -axis, which is 6.

11. The area of the triangle APQ can be calculated as:

$$\frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 2p \times 6 = 6p$$